# Harshitha Sai Maddukuri

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## SUMMARY

MS Computer Science student at Northeastern University graduating May 2027. Worked across the full development lifecycle on a production client application at Virtusa, then designed and shipped a full-stack app with real users and experimented hands-on with NLP and LLM systems. Comfortable with ambiguity and cross-team work. Authorized to work in the U.S. under F-1 OPT after graduation.

## TECHNICAL SKILLS

**Languages:** Java, C#, Python, TypeScript, JavaScript, SQL, HTML5, CSS3

**Frontend:** React, Next.js, Tailwind CSS, component architecture, async state management, server-side rendering, responsive design

**Backend:** Node.js, Django, .NET, RESTful APIs, OOP, JSON, SDLC

**AI & Machine Learning:** LLMs, prompt engineering, scikit-learn, NLP, pandas, NumPy, TF-IDF, SMOTE, model evaluation

**Databases:** PostgreSQL, MongoDB, MySQL, SQL schema design, query optimization

**DevOps & Tools:** Git, GitHub, CI/CD, Linux, Docker, Azure, Vercel, unit testing, code reviews, Agile/Scrum, technical documentation

## EXPERIENCE

**Software Engineer Intern – IpserLab, LLC (Remote)** **Sep 2026 – Dec 2026**

- Work on programming, design, testing, and ideation for a software product built with Java, JDBC, PostgreSQL, and React.
- Handle backend data access and persistence alongside the React frontend, with direct mentorship from senior engineers.

**Software Engineer Intern – Virtusa (India)** **Dec 2022 – Apr 2023**

- Owned features from design through testing and production deployment, building backend services and REST APIs in C# and .NET for a client application with active users.
- Wrote unit tests for every feature before it shipped and reviewed teammates' code in two-week Agile sprints, helping keep the codebase clean and maintainable.
- Traced production bugs to their root cause across development and test environments, and kept technical docs up to date.

## PROJECTS

**Snippr – Full-Stack Code Snippet Manager** (Next.js, TypeScript, React, Node.js, PostgreSQL, Supabase, Tailwind CSS, Vercel)

- Designed the PostgreSQL schema from scratch, built a type-safe REST API with secure authentication, and shipped to production on Vercel with automated CI/CD.
- Found a critical schema flaw through real user behavior; refactored the data layer and retested every affected route before redeploying, so no bad data reached users.
- Built server-side rendering, real-time filtering, and responsive layouts with Next.js and Tailwind CSS that stay fast on any device.

**Mental Health Detection from Reddit Posts** (Python, scikit-learn, NLTK, pandas, NumPy, TF-IDF, SMOTE)

- Built an NLP classification pipeline for 5,957 Reddit posts, testing 60 configurations of TF-IDF, SMOTE balancing, and five classifiers.
- Caught a label leakage bug mid-run and reran the full pipeline before reporting results; the tuned model reached 84.98% accuracy and 0.8371 macro F1 with stable five-fold cross-validation.

**Kambaz – Learning Management System** (React, TypeScript, Node.js, MongoDB, PostgreSQL, SQL, Git)

- Built backend REST APIs with role-based access control and designed the SQL schema for courses, assignments, enrollments, and grades across student, instructor, and admin roles.
- Wrote unit tests before each service shipped, reviewed teammates' code every sprint, and kept API documentation current so the frontend team could integrate without guesswork.

**Signature Verification using Siamese Neural Networks** (Python, Django, Java, HTML, CSS, Deep Learning)

- Built the testing pipeline for an AI identity verification system: image preprocessing, a Siamese neural network trained with contrastive loss, and a Django REST backend with a browser interface.
- Tuned false accept and false reject thresholds for reliable results, and published the full methodology in the IJFANS Journal.

## EDUCATION

**Northeastern University** – Boston, MA **Jan 2025 – May 2027 (Expected)**
**MS in Computer Science, Khoury College of Computer Sciences**
Coursework: Algorithms, Machine Learning, Cyber Security, DBMS, Web Development, Program Design Paradigms, HCI

**Vasireddy Venkatadri International Technological University** – India **Jun 2019 – Jun 2023**
**B.Tech in Computer Science and Engineering**
Coursework: Big Data Analysis, Unix and Shell, Java, ML, AI, Data Structures, Discrete and Engineering Mathematics

**Certifications:** IBM Full Stack Developer, Microsoft AZ-900, Python for Everybody (UMich), AI for Everybody, Anthropic AI Fluency
**Languages:** Telugu, Hindi, English (fluent)